Symmetry Method Homework 5

1). Pseudo Orthogonal $O(1,1)$

Consider $O(1,1)$ group parametrized by

$$\Lambda(\eta) = \begin{pmatrix} \cosh\eta & \sinh\eta \\ \sinh\eta & \cosh\eta \end{pmatrix}$$

1.1) Prove that $\Lambda(\eta)$ satisfies pseudo orthogonal condition

$$\Lambda^T(\eta)\, G\, \Lambda(\eta) = G$$

1.2) Find the generators of this group

2). Diagonal subgroup of $SU(2)$

Consider a subgroup of $SU(2)$ parametrized by

$$U(\theta) = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

2.1) Prove that $U(\theta)$ satisfies unitary condition

and $\det U(\theta) = 1$, i.e., $U(\theta) \in SU(2)$

2.2) Find the generators of this group

3). Algebra of $SO(3)$

Consider 3 generators of $SO(3)$:

$$\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} = \left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\}$$

Find all components of the structure constant

4). Traceless generators

Consider 3 traceless generators :

$$\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}$$

4.1) Find all components of the structure constant

4.2) Compute the group generated by $\hat{X}_1$

4.3) Compute the group generated by $\hat{X}_2$

4.4) Compute the group generated by $\hat{X}_3$

5) Echelon generators

Consider 3 generators :

$$\{\hat{X}_1, \hat{X}_2, \hat{X}_3\} = \left\{ \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \right\}$$

5.1) Find all components of the structure constant

5.2) Compute the group generated by $\hat{X}_1$

5.3) Compute the group generated by $\hat{X}_2$

5.4) Compute the group generated by $\hat{X}_3$